

# The stuff your degree assumed you already knew

One page. The networking words you recognise but couldn't explain yet — in plain English, with the picture that makes each one click.



A network is just computers trying to talk to each other. Everything below is a detail of *how* they pull that off.

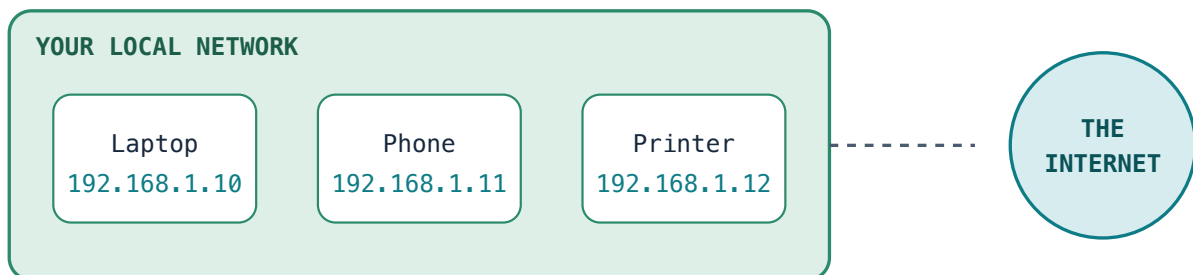
## PART 1 · THE ADDRESS SYSTEM

### IP address

A number that says **where a device is** on the network, so messages know where to go and how to get back.

#### THINK OF IT AS

A house address. No address, no mail.



Each device gets its own address inside the local network.

### Private vs public IP

**Private** = your address inside your own network (everyone's network reuses the same ranges, like 192.168.x.x). **Public** = the single address the outside world sees for your whole network.

#### THINK OF IT AS

Private = your room number inside a hotel. Public = the hotel's street address that couriers actually deliver to.

## Subnet mask

It draws the line in an IP address between "**which part is the neighbourhood**" and "which part is the specific house." That's how a device instantly knows if another device is **local** (talk directly) or **far away** (send it via the gateway).

### THINK OF IT AS

The line that splits a postal code from the house number. It decides who counts as your neighbour.

## Default gateway

The one address a device sends traffic to when the destination is **not** on the local network. Usually your router.

### THINK OF IT AS

The exit door out of your neighbourhood. Anything not local goes through that door to reach the wider world.

## MAC address

A **permanent** hardware ID burned into each network chip. An IP can change depending on the network you join; the MAC stays the same.

### THINK OF IT AS

Your passport number (never changes) vs your current home address, the IP (changes when you move).

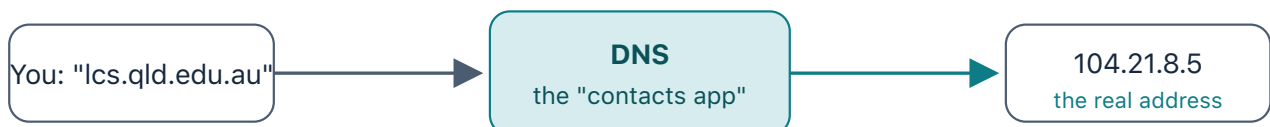
## PART 2 · FINDING THINGS

## DNS

The system that turns a **name** humans remember (google.com) into the **number** machines need (the IP). You type a name; DNS hands back the address.

### THINK OF IT AS

The contacts app on your phone. You tap "Mum," it dials the actual number. You don't memorise the digits.



Name in → address out. That's the whole job of DNS.

## Ports

One device runs many services at once, so each gets a numbered **door**. Web traffic uses 443 (secure) and 80; email, file shares, etc. each have their own.

### THINK OF IT AS

An apartment block at one street address — the port number is the apartment you're actually visiting.

## PART 3 · THE BOXES THAT MOVE TRAFFIC

## Switch vs router

A **switch** connects devices **within one** network and passes traffic between them. A **router** connects **different networks** together — most importantly, yours to the internet.

### THINK OF IT AS

The switch is the hallway connecting rooms in one building. The router is the front gate connecting your building to the whole city.

## PART 4 · KEEPING IT SAFE

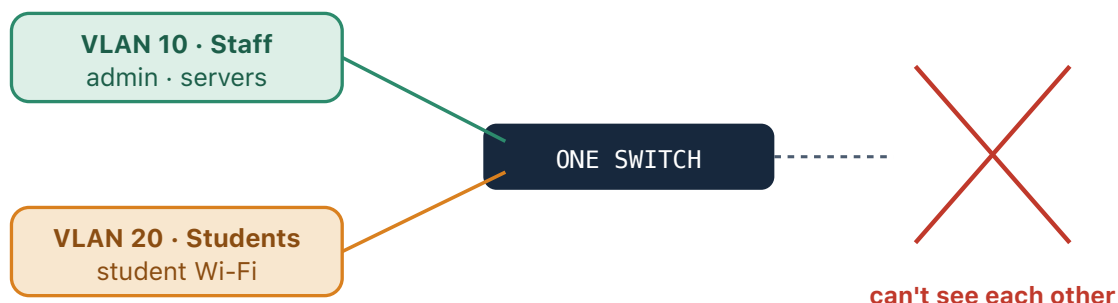
## VLAN

A way to split **one physical network into several separate logical ones**, so groups can't see each other even though they share the same switches and cables.

### THINK OF IT AS

Invisible internal walls. Same building, but the student wing can't wander into the staff offices.

**LCS** This is exactly Chris's **relief-teacher / student-vs-staff access** problem. "Student Wi-Fi shouldn't see the staff admin servers" = put them on different VLANs.



Same hardware, walled into separate networks.

## Firewall

A checkpoint that decides which traffic is **allowed in or out** based on rules. Blocks the unwanted, lets the approved through.

### THINK OF IT AS

The bouncer at the door checking everyone against the list before they get in.

**LCS** The school's **content filtering** — stopping students reaching inappropriate sites — lives at this layer. Same idea: rules deciding what traffic is allowed.

### PUT IT TOGETHER

## What actually happens when you open a website

Every term above, in the order it fires. Read this once and the whole map locks in.

- 1 You type **lcs.qld.edu.au**. Your device asks **DNS** "what's the address?" and gets back an IP.
- 2 Your device checks its **subnet mask**: is that IP local or far away? It's far away.
- 3 So it hands the request to the **default gateway** (your router) — the exit door.
- 4 The **router** sends it out across the internet, hopping between networks until it reaches the right server.
- 5 A **firewall** along the way checks the traffic is allowed. It knocks on the right **port** (443 for a secure site).
- 6 The server answers, the page travels all the way back to **your IP address** — and loads. All of that in under a second.